

ALFIYA ANSARI

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SUMMARY

Data Science student with the hands-on experience in data analysis, machine learning, and statistical modeling using Python and SQL. Strong foundation in exploratory data analysis, feature engineering, and predictive modeling on real-world datasets. Experienced in transforming raw data into actionable insights through structured analytical workflows. Effective team collaborator with the ability to communicate data-driven insights clearly.

EXPERIENCE

Data Science Intern – AB Technologies

Feb 2023 – Mar 2023

Nagercoil, India

- Built end-to-end data pipelines using Python, SQL, MySQL, and MongoDB, leveraging Pandas and NumPy for data cleaning, pre-processing, and exploratory data analysis (EDA) on 100K+ structured datasets.
- Applied statistical analysis, hypothesis testing, regression models, and classification models to extract business insights and support KPI analysis and reporting.
- Developed and optimized machine learning and deep learning models using scikit-learn and TensorFlow, implementing feature engineering techniques to improve predictive performance by 50%+.
- Created interactive dashboards and visualizations using Tableau, Power BI, Excel, Matplotlib, and Seaborn to communicate analytical findings and enable data-driven decision-making.

Volunteer Tutor – RCET

Sep 2022 – Mar 2023

Kanyakumari, India

- Delivered structured training in Python, data analysis, data science fundamentals, and statistical analysis, guiding students through exploratory data analysis, data cleaning workflows, and model development.
- Mentored students in applying machine learning, deep learning, regression, and classification techniques to real-world datasets while strengthening their analytical thinking and reporting skills.

Research Assistant (AI & Data Science) – University of Koblenz

Oct 2025 – Present

Koblenz, Germany

- Conducted comparative research on supervised machine learning and deep learning approaches for structured tabular classification and multivariate time series forecasting.
- Implemented models including , Random Forest, XGBoost, and Neural Networks using Python, scikit-learn, TensorFlow, and PySpark, evaluating performance with ROC-AUC, RMSE, and other statistical metrics.
- Applied advanced feature engineering, model validation, statistical testing, and predictive analytics techniques to ensure robust and scalable analytical solutions.

EDUCATION

Masters in Web and Data Science

University of Koblenz | Current CGPA: 2.2

Oct 2024 – Present

Bachelors in Computer Science

Anna University | CGPA: 1.7

Aug 2019 – Jul 2023

CERTIFICATIONS

- Google Data Analytics Certificate: Coursera
- Data Science Bootcamp: Udemy
- Machine Learning with Python: Udemy
- Deep Learning Specialization: DeepLearning.AI
- SQL for Data Analytics: Udemy

SKILLS

Python SQL MongoDB
Data Analysis Data Science
Tableau Power BI Excel
Data Cleaning Statistical Analysis
Exploratory Data Analysis
Pandas Numpy Seaborn
Hypothesis Testing KPI Reporting
Dashboard Development PySpark
Machine Learning Deep Learning
LLM Amazon Web Service
M365 Git Docker
Business Insights

PROJECTS

Bayut (UAE) Real Estate Market Intelligence Dashboard 🔗

- Cleaned and transformed 41,000 UAE property listings using SQL and Python, removing high-null columns (60%) and engineering structured pricing features.
- Developed an interactive Tableau dashboard for business intelligence reporting, analyzing AED 156.7B total sales value and average property price (AED 3.79M).
- Identified KPI trends including a 2022 price peak (AED 13.6M average) and luxury dominance in penthouse segments (AED 23.46M avg).
- Conducted city-level comparative analysis to support data-driven market insights and reporting.
- **Tools:** SQL, Excel, Python, Pandas, Tableau, etc.

Quantile-Based Real Estate Valuation & Overpricing Detection System 🔗

- Designed a quantile-based price prediction framework using LightGBM and XGBoost to estimate fair market value and detect overpricing patterns.
- Built preprocessing pipelines with imputation, scaling, and one-hot encoding, generating 231 engineered features.
- Evaluated models using 5-fold TimeSeriesSplit cross-validation; achieved MAE of 189,153 (XGBoost) and 193,210 (LightGBM).
- Constructed prediction intervals (P10–P90) with 99.78% coverage for uncertainty aware pricing analysis.
- **Tools:** Python, Pandas, NumPy, Scikit-learn, LightGBM, XGBoost, etc.

Healthcare Admission Forecasting & Clinical Risk Analytics 🔗

- Developed an end-to-end healthcare analytics pipeline for hospital admission forecasting and operational capacity planning.
- Performed exploratory data analysis (EDA) and engineered structured features (LOS category, ICU requirement, temporal attributes).
- Compared Logistic Regression, Random Forest, and XGBoost models, achieving 68% accuracy on multiclass LOS prediction (3,152 test samples).
- Deployed a Streamlit-based dashboard for interactive analytics and patient level risk assessment.
- **Tools:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Streamlit, etc.

Distributed Fraud Risk Classification using PySpark 🔗

- Built a scalable distributed data pipeline using PySpark for preprocessing, feature encoding, and vector assembly.
- Implemented a Logistic Regression classifier using Spark MLlib for fraud risk prediction.
- Achieved 98.7% accuracy (class imbalance observed), with 1.00 precision and 0.987 recall; evaluated using ROC-AUC (0.5713) and confusion matrix analysis.
- Developed modular Spark ML pipelines integrating StringIndexer, OneHotEncoder, and VectorAssembler.
- **Tools:** PySpark, Python, Pandas, Seaborn, etc.

LANGUAGES

English — C1 (Advanced)

German — Beginner

COMPETENCIES

- Data analysis, data validation, and data quality assurance
- Data cleaning, pre-processing, and exploratory data analysis (EDA)
- SQL-based data querying, aggregation, and relational database handling
- Complete Statistical analysis and hypothesis testing for data-driven decision making
- Business intelligence reporting, KPI tracking, and dashboard development
- Data visualization and stakeholder oriented reporting

ACHIEVEMENTS

- Conducted academic research in applied machine learning and time-series forecasting during Master's studies at University of Koblenz.
- Academic Excellence Award (3×) – Graduated with CGPA 9.0/10 (First Class with Distinction).
- Winner and finalist in inter-college and national-level technical competitions.
- Elected Council Member and contributing to academic and technical event coordination.

STRENGTHS

- Analytical thinking
- Problem solving
- Data-driven mindset
- Adaptability
- Continuous learning
- Teamwork
- Attention to detail
- Critical thinking